What is No Code Algo Trading

NON TECH TRADING

Stocks, Futures & Options Trading

All in one trading platform - Trade with advanced tools like multi-leg strategy builder, Tick-live mark...

Nubra Trading Platform

No Code Algo Trading

Understand No Code Algo Trading that opens doors to all

doesn’t require algorithm knowledge, beneficial for non technical traders and stakeholders

@reallygreatsite.com

Stocks, Futures & Options Trading

All in one trading platform - Trade with advanced tools like multi-leg strategy builder, Tick-live mark...

Nubra Trading Platform

Benefits of No Code Algo Trading

- Increased Accessibility
- Faster Strategy Development
- Cost and Time Savings
- Improved Efficiency

No Code reduces time spent on technical depth

@reallygreatsite.com

Stocks, Futures & Options Trading

All in one trading platform - Trade with advanced tools like multi-leg strategy builder, Tick-live mark...

Nubra Trading Platform

Platforms for Pre Written Algo for Trading

uTrade Algos
Kuants
RoboMatic
Nubra

Advantages like drag-and-drop tools, pre-built templates, and visual strategy builders

@reallygreatsite.com

Want more Algo Trading tips?

Follow us and share this post with others who might find it helpful!

LINKEDIN
@reallygreatsite.com

Backtesting and EMA Strategy

What is Backtesting?

Backtesting is a process used in trading and investing to test how a trading strategy would have performed in the past using historical market data. Instead of risking real money, the strategy is simulated on past prices to see whether it would have been profitable or not.

The key idea is:

- If a strategy performs well on past data, it **may** perform well in the future.
- Backtesting helps traders **evaluate profitability, risk, and consistency** of a strategy before applying it in live markets.

Commonly evaluated metrics in backtesting:

- **Total Return** – how much profit/loss the strategy made.
- **Sharpe Ratio** – risk-adjusted return.
- **Win Rate** – percentage of profitable trades.
- **Drawdown** – the largest drop from a peak to a low (risk measure).

Backtesting with Exponential Moving Averages (EMA)

An **Exponential Moving Average (EMA)** is a type of moving average that gives **more weight to recent prices**, making it more responsive to market changes compared to a Simple Moving Average (SMA).

In this strategy:

- We use **two EMAs** – a short-term EMA (20 periods) and a long-term EMA (50 periods).
- **Buy Signal**: When the short EMA crosses **above** the long EMA → market trend turning bullish.
- **Sell Signal**: When the short EMA crosses **below** the long EMA → market trend turning bearish.

This is called a **crossover strategy**, widely used to capture medium-term market trends.

PYTHON CODE AND OUTPUT FOR BACKTESTING

```
from backtesting import Backtest, Strategy
from backtesting.lib import crossover
from backtesting.test import GOOG   #Google Stock Data
import pandas as pd

def EMA(series, period):
    return pd.Series(series).ewm(span=period, adjust=False).mean()

class EmaCross(Strategy):
    n1 = 20    # Short-term EMA
    n2 = 50    # Long-term EMA

    def init(self):
        close = self.data.Close
        self.ema1 = self.I(EMA, close, self.n1)  # Short EMA
        self.ema2 = self.I(EMA, close, self.n2)  # Long EMA

    def next(self):
        # Buy when short EMA crosses above long EMA
        if crossover(self.ema1, self.ema2):
            self.position.close()
            self.buy()
        # Sell when short EMA crosses below long EMA
        elif crossover(self.ema2, self.ema1):
            self.position.close()
            self.sell()


bt = Backtest(GOOG, EmaCross,
              cash=10000, commission=.002,
              exclusive_orders=True)

output = bt.run()
print(output)
bt.plot()
```

```
Start                     2004-08-19 00:00:00
End                       2013-03-01 00:00:00
Duration                   3116 days 00:00:00
Exposure Time [%]                    91.06145
Equity Final [$]                  14649.35996
Equity Peak [$]                   27874.16682
Commissions [$]                    2131.90204
Return [%]                            46.4936
Buy & Hold Return [%]               703.45824
Return (Ann.) [%]                     4.58119
Volatility (Ann.) [%]                29.19315
CAGR [%]                              3.13599
Sharpe Ratio                          0.15693
Sortino Ratio                         0.23584
Calmar Ratio                          0.08201
Alpha [%]                            14.42932
Beta                                  0.04558
Max. Drawdown [%]                    -55.8585
Avg. Drawdown [%]                     -6.5942
Max. Drawdown Duration     1152 days 00:00:00
Avg. Drawdown Duration       79 days 00:00:00
# Trades                                   33
Win Rate [%]                         30.30303
Best Trade [%]                       91.74694
Worst Trade [%]                      -17.9203
Avg. Trade [%]                        0.79261
Max. Trade Duration         309 days 00:00:00
Avg. Trade Duration          86 days 00:00:00
Profit Factor                         1.47785
Expectancy [%]                         2.5128
SQN                                   0.17876
Kelly Criterion                       0.02647
_strategy                            EmaCross
_equity_curve                             ...
_trades                          Size  EntryB...
dtype: object
```

ALGO TRADING: EXPONENTIAL MOVING AVERAGES

Newsletter

September 2025 Edition

EXPONENTIAL MOVING AVERAGES

The Exponential Moving Average (EMA) is a line drawn on a price chart that helps traders see the overall direction of the market.

Unlike a Simple Moving Average (SMA), which treats all past prices equally, the EMA pays more attention to recent prices. This makes it react faster to new price changes.

- If prices suddenly go up, the EMA line will rise more quickly than the SMA.
- If prices fall, the EMA will drop sooner, helping traders spot changes faster.

👉 Think of it like this:

- SMA = a teacher who treats all students equally
- EMA = a teacher who listens more to the most recent answers in class

That's why traders like EMAs for fast-moving markets like stocks, forex, or crypto — they catch trends earlier than SMAs.

HOW IT WORKS

- Weighting Recent Data: EMA prioritizes the latest prices, reducing lag in signal generation.
- Trading Signals:
  - When the price rises above EMA, it often indicates a buy (long) signal.
  - When the price falls below EMA, it often indicates a sell (short) signal.
- Flexibility: EMAs can be applied to any timeframe, from 1-minute crypto charts to 200-day equity charts.

FAST EDDIE'S BOT

The Eddie Bot, when trained on Simple Moving Averages, was missing on opportunities because by the time it showed the trend, the opportunity was already gone because of slow aggregation over past and present data. The EMA's reduced lag time meant Eddie ' s algorithm could enter and exit positions faster, capturing more profits in fastchanging markets. Instead of waiting for a slow signal, Eddie ' s bot was already acting on new information

9861912635

https://www.linkedin.com/in/profpiyusharora/

piyusharora.epsilon@gmail.com

Advantages of EMA

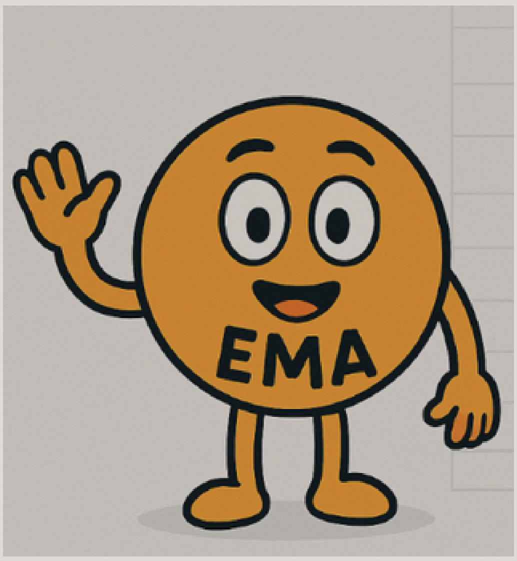
- Reacts faster to price changes than SMA.
- Great for volatile markets (crypto, forex).
- Backbone of popular indicators (MACD, Bollinger Bands).
- Works across all timeframes (1-min to daily).

Limitations of EMA

- Too sensitive in sideways markets → false signals.
- Requires confirmation with RSI/Volume.
- Works best when market has a clear trend.

Pro Tips for Traders

- Use dual EMAs (short-term + long-term).
- Adjust EMA length depending on volatility.
- Combine EMA signals with candlestick confirmations.



Trend Detection

EMAs help trading bots identify whether the ongoing ttrend is bullish (increasing) or bearish (decreasing).

Entry & Exit Signals

- Crossover strategies (e.g., 9-day crossing 21-day).
- Helps traders catch breakouts earlier than SMA.

Dynamic Support & Resistance

- EMA acts as a “moving floor/ceiling.”
- Traders buy near EMA support during uptrends and sell at EMA resistance in downtrends.

ALGO TRADING WITH EXPONENTIAL MOVING AVERAGES

Issue 1 September 2025

THE POWER OF EXPONENTIAL MOVING AVERAGES IN ALGO TRADING

Proof of Concept: EMA in Action

Case Study: NASDAQ 100 (2019-2023)

- EMA crossover strategy outperformed SMA by +8% annualized returns.
- Worked best in strong trending markets.
- Struggled in sideways conditions → false signals.

9861912635

https://www.linkedin.com/in/profpiyusharora/

piyusharora.epsilon@gmail.com